

# INFO8006 Introduction to Artificial Intelligence

## Exercise session 3: Quantifying uncertainty

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### RECAP

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#### Definitions

A probability is a degree of belief in a proposition.

|                          |                                                                                               |
|--------------------------|-----------------------------------------------------------------------------------------------|
| Sample space             | $\Omega$ , of possible worlds $\omega$                                                        |
| Axioms                   | $0 \leq P(\omega) \leq 1$ , $\sum_{\omega \in \Omega} P(\omega) = 1$                          |
| Event                    | $A \subseteq \Omega$ , $P(A) = \sum_{\omega \in A} P(\omega)$                                 |
| Random variable          | $X : \Omega \rightarrow D_X$ , with $X = x$ the event $\{\omega \in \Omega : X(\omega) = x\}$ |
| Distribution             | $\mathbf{P}(X)$ , the numbers $P(x) = P(X = x)$ for $x \in D_X$                               |
| Joint distribution       | $\mathbf{P}(X_1, \dots, X_n)$ , the numbers $P(x_1, \dots, x_n)$                              |
| Conditional probability  | $P(a   b) = P(a, b) / P(b)$ , for $P(b) > 0$                                                  |
| Conditional distribution | $\mathbf{P}(X   Y)$ , one $\mathbf{P}(X   y)$ for each $y \in D_Y$                            |

Random variables are written in upper case and their values in lower case; a boolean variable  $A$  takes the values  $a$  and  $\neg a$ .  $P(x)$  is a number,  $\mathbf{P}(X)$  a table.

#### Rules

|                                           |                                               |
|-------------------------------------------|-----------------------------------------------|
| Complement                                | $P(\neg a) = 1 - P(a)$                        |
| Union                                     | $P(a \vee b) = P(a) + P(b) - P(a \wedge b)$   |
| Product rule                              | $P(a, b) = P(a   b)P(b) = P(b   a)P(a)$       |
| Sum rule, total probability               | $P(x) = \sum_y P(x, y) = \sum_y P(x   y)P(y)$ |
| Chain rule                                | $P(x_{1:n}) = \prod_i P(x_i   x_{1:i-1})$     |
| Bayes' rule                               | $P(a   b) = P(b   a)P(a) / P(b)$              |
| Independence, $X \perp Y$                 | $P(x, y) = P(x)P(y)$                          |
| Conditional independence, $X \perp Y   Z$ | $P(x, y   z) = P(x   z)P(y   z)$              |
| Naive Bayes                               | $P(c, e_{1:n}) = P(c) \prod_i P(e_i   c)$     |

The rules hold for all values, and between distributions. Naive Bayes assumes the effects  $E_{1:n}$  to be conditionally independent given the cause  $C$ . The joint distribution answers every query: select the entries consistent with the evidence, sum out the other variables, normalize.

#### Bayes' rule

A model is given in the *causal* direction: for an agent, a *prior*  $\mathbf{P}(S)$  over the unobserved state and a *sensor model* (or *likelihood*)  $\mathbf{P}(e | S)$  of its percepts. The *diagnostic* question is answered by the *belief*, or *posterior*,  $\mathbf{P}(S | e) = \frac{1}{Z} \mathbf{P}(e | S) \mathbf{P}(S)$ , where  $Z = P(e) = \sum_s P(e | s)P(s)$  is the probability of the *evidence*. For percepts conditionally independent given the state, each belief is the prior of the next,  $\mathbf{P}(S | e_{1:k+1}) = \frac{1}{Z} \mathbf{P}(e_{k+1} | S) \mathbf{P}(S | e_{1:k})$ .

*In session:* exercises 1, 2, 3, 5.

**Exercise 1 Rational beliefs** (AIMA ch. 12)

Let  $A, B \subseteq \Omega$  be two *events*. An agent holds the beliefs  $P(A) = 0.4$  and  $P(B) = 0.3$ . What ranges of probabilities would it be rational for the agent to hold for the events  $A \cup B$  and  $A \cap B$ ? What if the agent also believes that  $P(A \mid B) = 0.5$ ?

**Exercise 2 Full joint distribution** (AIMA ch. 12)

A monitoring program supervises the bearings of a wind turbine. A worn bearing (*fault*) tends to make the turbine vibrate (*vibration*) and to trigger the alarm of a diagnostic probe (*alarm*). The program holds the joint distribution  $\mathbf{P}(V, A, F)$  below, over the boolean random variables  $V$  (vibration),  $A$  (alarm) and  $F$  (fault). Compute the following probabilities.

|          | $v$   |          | $\neg v$ |          |
|----------|-------|----------|----------|----------|
|          | $a$   | $\neg a$ | $a$      | $\neg a$ |
| $f$      | 0.108 | 0.012    | 0.072    | 0.008    |
| $\neg f$ | 0.016 | 0.064    | 0.144    | 0.576    |

- (a)  $P(v)$
- (b)  $P(f)$
- (c)  $P(v \mid f)$
- (d)  $P(f \mid v \vee a)$
- (e) Are  $V$  and  $A$  independent? Are they conditionally independent given  $F$ ?

**Exercise 3 Fraud detection** (AIMA ch. 12)

A fraud detection program screens card payments. It is 99 % accurate, *i.e.* it flags a fraudulent payment with probability 0.99 and a legitimate payment with probability 0.01. Fraud is rare: it strikes only 1 payment in 10 000. The program has just flagged your payment. Why is it good news, for you, that fraud is rare? What is the probability that the payment is actually fraudulent?

**Exercise 4 Conditional independence** (AIMA ch. 12)

For each of the following statements, about values  $a, b, c$  of the random variables  $A, B, C$ , either prove it is valid or give a counterexample. Conditional probabilities are only taken given events of positive probability, as their definition requires.

- (a) If  $P(a, b, c) > 0$  and  $P(a \mid b, c) = P(b \mid a, c)$ , then  $P(a \mid c) = P(b \mid c)$ .
- (b) If  $P(a \mid b, c) = P(a)$ , then  $P(b \mid c) = P(b)$ .
- (c) If  $P(a \mid b) = P(a)$ , then  $P(a \mid b, c) = P(a \mid c)$ .

**Exercise 5 Driving styles** (AIMA ch. 13)

A self-driving car shares the highway with another car, whose driving style it does not know. It models three styles, cautious ( $a$ ), normal ( $b$ ) and aggressive ( $c$ ), equally likely a priori, under which the other car changes lanes during a given minute with probabilities 0.2, 0.6 and 0.8, respectively, independently from one minute to the next given its style. The self-driving car observes the other car for three minutes. Let  $Y \in \{a, b, c\}$  be the style of the other car, and  $X_i \in \{l, k\}$  tell whether it changes lanes ( $l$ ) or keeps its lane ( $k$ ) during minute  $i$ .

- (a) Write the joint distribution  $\mathbf{P}(Y, X_1, X_2, X_3)$  of this model as a product of smaller tables, and give these tables.

- (b) Determine the most likely driving style if the other car changed lanes during the first two minutes and kept its lane during the third.
- (c) The car updates its belief after each minute, the belief after one minute being the prior of the next. Compute its belief after one, two and three minutes, and compare with (b).

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QUIZ

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1. Bayes' rule states that  $P(y \mid x)$  is equal to ...

- ☐  $\frac{P(x \mid y)P(x)}{P(y)}$ .
- ☐  $\frac{P(x \mid y)P(y)}{P(x)}$ .
- ☐  $\frac{P(x, y)}{P(y)}$ .
- ☐  $P(x \mid y)P(y)$ .

2. Which of the following is true?

- ☐  $P(x, y \mid z) = \frac{P(x \mid y, z)}{P(y, z \mid x)}$ .
- ☐  $P(x, y, z) = P(x)P(y)P(z)$ .
- ☐  $P(x, y \mid z) = P(y \mid x, z)P(x \mid z)$ .
- ☐  $P(x \mid y, z) = \frac{P(x, y, z)}{P(y)P(z)}$ .

3. The naive Bayes model assumes ...

- ☐ conditional independence between the effects, given the cause.
- ☐ independence between the effects.
- ☐ independence between the effects and the cause.
- ☐ conditional independence between each effect and the cause, given the other effects.

4. With a uniform prior  $\mathbf{P}(A)$ , the posterior  $\mathbf{P}(A \mid b)$  is proportional to ...

- ☐ the prior  $\mathbf{P}(A)$ .
- ☐ the likelihood  $\mathbf{P}(b \mid A)$ .
- ☐ the evidence  $P(b)$ .
- ☐ the marginal  $\mathbf{P}(B)$ .